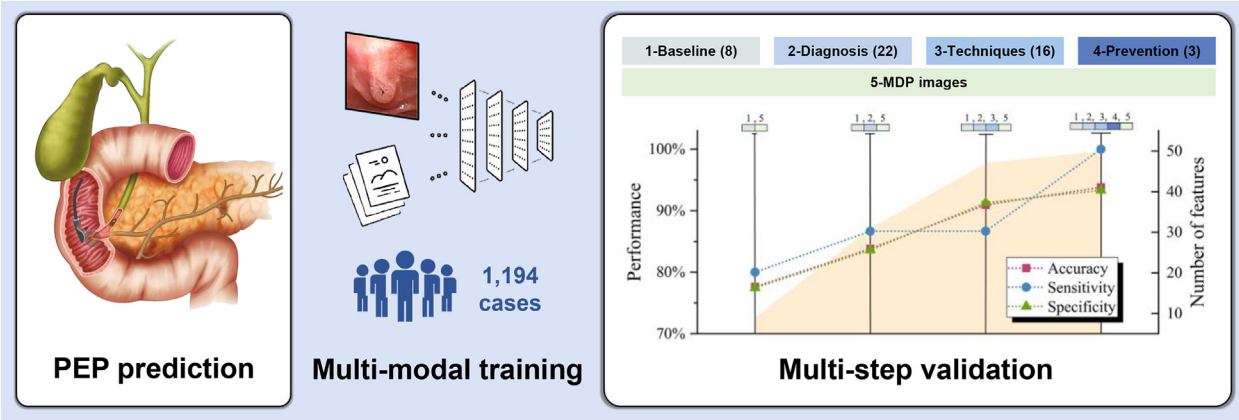
Multistep validation of a post-ERCP pancreatitis prediction system integrating multimodal data: a multicenter study

Youming Xu, MD,[1,2,*] Zehua Dong, MD,[1,2,*] Li Huang, MM,[1,2] Hongliu Du, MD,[1,2] Ting Yang, MD,[1,2] Chaijie Luo, MD,[1,2] Xiao Tao, MM,[1,2] Junxiao Wang, MM,[1,2] Zhifeng Wu, MM,[1,2] Lianlian Wu, MD,[1,2] Rong Lin, MD, PhD,[3] Honggang Yu, MD, PhD[1,2]

Wuhan, China

GRAPHICAL ABSTRACT



Background and Aims: The impact of various categories of information on the prediction of post-ERCP pancreatitis (PEP) remains uncertain. We comprehensively investigated the risk factors associated with PEP by constructing and validating a model incorporating multimodal data through multiple steps.

Methods: Cases (n = 1916) of ERCP were retrospectively collected from multiple centers for model construction. Through literature research, 49 electronic health record (EHR) features and 1 image feature related to PEP were identified. The EHR features were categorized into baseline, diagnosis, technique, and prevention strategies, covering pre-ERCP, intra-ERCP, and peri-ERCP phases. We first incrementally constructed models 1 to 4 incorporating these 4 feature categories and then added the image feature into models 1 to 4 and developed models 5 to 8. All models underwent testing and comparison using both internal and external test sets. Once the optimal model was selected, we conducted comparisons among multiple machine learning algorithms.

Results: Compared with model 2 that incorporated baseline and diagnosis features, adding technique and prevention strategies (model 4) greatly improved the sensitivity (63.89% vs 83.33%, $P < .05$) and specificity (75.00% vs 85.92%, $P < .001$). A similar tendency was observed in the internal and external tests. In model 4, the top 3 features ranked by weight were previous pancreatitis, nonsteroidal anti-inflammatory drug use, and difficult cannulation. The image-based feature has the highest weight in models 5 to 8. Finally, model 8 used a random forest algorithm and showed the best performance.

Conclusions: We first developed a multimodal prediction model for identifying PEP with a clinical-acceptable performance. The image and technique features are crucial for PEP prediction. (Gastrointest Endosc 2024;100:464-72.)

(footnotes appear on last page of article)

ERCP is an interventional endoscopic procedure widely used in the management of pancreaticobiliary disorders.[1] Post-ERCP pancreatitis (PEP) is the most frequent and significant among the various adverse events associated with ERCPs, which is estimated to occur in approximately 8% of average-risk procedures and 15% of high-risk procedures.[2] Notably, between 2011 and 2017, an increase in the overall incidence rate of PEP-related hospital admissions rose from 4.29% to 5.22%.[3] Additionally, PEP-related mortality also saw an increase during this period, climbing from 2.8% to 4.4%.[4]

Understanding the risk factors associated with PEP is crucial for healthcare providers to enhance medical care and prevention strategies.[1,5] Over the past few decades, numerous studies have identified several factors that may contribute to the development of PEP, including sphincter of Oddi dysfunction, female sex, age <60 years, normal bilirubin, difficult cannulation, and a pancreatic duct injected or deeply accessed with a guidewire.[2,6-19] Based on these factors, several risk-prediction models have been developed using various methods.[20-26] However, most of these studies have primarily relied on data from the electronic health record (EHR), neglecting to consider the images captured during the ERCP procedure. Interestingly, research has indicated a correlation between the likelihood of PEP and the macroscopic shape of the major duodenal papilla (MDP).[27] Consequently, it is valuable to incorporate this factor into the prediction process.

In addition, most studies focusing on PEP risk prediction have primarily analyzed patient-related features and procedure duration while overlooking the inclusion of manual factors.[20,22,24-26] Reports have shown that the PEP rate is significantly associated with the scale or expertise of the institution (eg, teaching hospitals tend to have a lower PEP rate compared with smaller institutions) and the procedural volume of the endoscopists.[3] However, these factors are broad in nature and may lack specific clinical guidance. They can be more precisely reflected by certain intraprocedure skills and operations. Therefore, it is imperative and worthwhile to deconstruct these factors into measurable metrics that can be artificially adjusted and controlled and subsequently incorporate them into the analysis.

Because of the neglect of image data and specific manual factors in previous studies, the contribution of these factors remains unclear. In this study, we developed a prediction model for risk stratification of PEP by integrating image data and EHR data, including MDP images and baseline, diagnostic, technique, and prevention information. To the best of our knowledge, this is the first study in PEP prediction to incorporate multimodal data and assess the relative contributions of factors through a multistep validation process.

METHODS

Data processing and description

A total of 1916 cases were included in the training and testing process. To comprehensively evaluate the structure and morphology of the MDP from different angles, we collected multiple images for each case. We manually constructed the 3 datasets.

Dataset 1 was the training set and comprised 1194 cases derived from 1168 patients who underwent ERCP at Renmin Hospital of Wuhan University, collected retrospectively. Specifically, of the 1194 cases, 82 were PEP cases with 367 MDP images and 1112 were cases without PEP accompanied by 5007 MDP images. Dataset 2 was the internal test set and comprised 512 cases from 508 patients who underwent ERCP at Renmin Hospital of Wuhan University, collected retrospectively. Thirty-six were PEP cases with 185 MDP images and 476 were cases without PEP accompanied by 1913 MDP images. Dataset 3 was the external test set and included 210 cases from 210 patients who underwent ERCP at Nanjing Drum Tower Hospital of Nanjing University Medical School and the Tenth People Hospital of Tongji University, collected retrospectively. This external test set comprised 15 cases of PEP with 52 MDP images and 195 non-PEP cases accompanied by 616 MDP images (Supplementary Table 1, available online at www.giejournal.org).

The inclusion criterion for cases were age >18 years and a naïve major papilla. The exclusion criteria for cases were patients with missing data, active or within 2 weeks before ERCP acute pancreatitis, and failed procedures. Images were taken that focused on the major papilla or near the major papilla before cannulation started (Supplementary Fig. 1, available online at www.giejournal.org). Eligible images of the MDP are shown in Supplementary Figure 2 (available online at www.giejournal.org).

Establishment of PEP risk factors

We determined the risk factors of PEP through literature research and complying with the International Classification of Diseases, 11th revision (ICD-11). We searched using the key words "Post-Endoscopic Retrograde Cholangiopancreatography Pancreatitis" OR "Post-ERCP Pancreatitis," "Risk factors" OR "feature" OR "characteristic," and "ERCP-Related Complications" OR "ERCP-Related Adverse Events" in the PubMed database between January 1, 2009 and May 31, 2023. One hundred twenty-five studies were assessed. Furthermore, 92 studies were excluded because they were unrelated to pancreatitis (n = 31), unrelated to risk factors of PEP (n = 56), and case reports (n = 5). One of 11 studies was not retrieved because the full text was unavailable. Then, 12 studies were assessed for eligibility; 7 were excluded because they were unrelated to PEP diagnosis features. Additionally, 8 records were added by a manual search. Finally, 15 studies were included.

Then, we finalized 49 EHR features with reference to the studies and ICD-11, which were subsequently categorized into 4 distinct sections: baseline, in which risk factors acquired before the ERCP procedure were determined by literature review; diagnosis, which was made by endoscopists during ERCP with reference to ICD-11 but before any direct or adjuvant treatment techniques were performed; techniques, which were manual factors such as endoscopist experience, difficult cannulation, and use of endoscopic sphincterotomy; and prevention strategies, which were finalized by the expert endoscopist according to existing ERCP-related treatment protocols, including the placement of pancreatic stents and use of nonsteroidal anti-inflammatory drugs (NSAIDs) and periprocedural fluid solutions. The full list of establishing features is shown in Supplementary Table 2 (available online at www.giejournal.org). Details of managing missing data are documented in Appendix 1 (available online at www.giejournal.org).

Development of prediction models

Five feature libraries within each dataset 1 to 3, encompassing the aforementioned 4 EHR sections and MDP images, were developed for training and testing (Fig. 1).

Feature extraction of MDP images

The MDP image feature was extracted using an independent deep convolutional neural network, specifically on dataset 1. Before training, all MDP images were first converted into a square template to isolate the central chroma region and then underwent contrast normalization. Then, we constructed a binary classification model (PEP or non-PEP) based on the ResNet-50 using 5374 preprocessed MDP images from 1194 cases. After this, the aforementioned image model was used to generate individualized risk scores based on the images for each case.

Development and assessment of EHR-only models

We trained 4 EHR-only models, namely model 1 to model 4, including 49 features extracted from the EHR and incorporating distinct sections in a sequential manner. These sections were baseline, diagnosis, techniques, and prevention strategies. Model 1 was trained using 8 baseline features before ERCP, model 2 was trained using 8 baseline and 22 diagnosis features, model 3 was trained using 46 features with the addition of 16 technique features, and model 4 was trained using all 49 features with the addition of 3 routine prevention strategies. The performance of EHR-only models were tested in datasets 2 and 3.

Development and assessment of fusion models

The image-based factor, which captured clinical features extracted from image data, was integrated with the 49 EHR features to derive a comprehensive probability. Based on the feature combinations of models 1 to 4, the image-based factor was included as an input for constructing models 5 to 8. The performance of the fusion models was evaluated using datasets 2 and 3.

Constructing and comparing multiple machine learning models

We applied the random forest algorithm to develop models 1 through 8. The best model among models 1 through 8 was selected based on the internal test set; we then conducted a comparison among multiple machine learning algorithms. The features of the optimal model were used for further training and comparison against another 6 machine learning algorithms: the Gaussian naïve Bayes, k-nearest neighbor, logistic regression, decision tree, support vector machine, and gradient-boosting decision tree. The algorithm that demonstrated superior performance was then selected as the ultimate best model.

Ethics

This study was approved by the Ethics Committee of Renmin Hospital of Wuhan University, Nanjing Drum Tower Hospital of Nanjing University Medical School, and the Tenth People Hospital of Tongji University (ethics no. WDRY2019-K073(X05). Informed consent was deemed exempt for the retrospectively collected data by the institutional review boards.

Statistical analysis

For evaluating the performance of EHR-only models and fusion models, we calculated the area under the receiver-operating characteristic curve (AUC), accuracy, sensitivity, and specificity for each classification threshold applied to the test datasets. The 95% confidence interval (CI) was determined by calculating the 2.5 and 97.5 percentiles of the empirical distribution of the corresponding metrics using the Wilson procedure. The McNemar test was used to compare the accuracy, sensitivity, and specificity. Comparison of demographic characteristics among the 3 hospitals was performed using analysis of variance and the χ^2 test for continuous and categorical variables, respectively. A $P < .05$ was considered statistically significant.

RESULTS

Demographic characteristics

The analysis of demographic characteristics of patients from multiple hospitals revealed no statistically significant difference between dataset 2 (hospital 1) and dataset 3 (hospitals 2 and 3), as determined by the χ^2 test (Supplementary Table 3, available online at www.giejournal.org).

Performance of EHR data–only models

The accuracy of EHR-only models 1 to 4 for comparing the model enhancement of baseline features, diagnosis features, technique features, and prevention strategies are

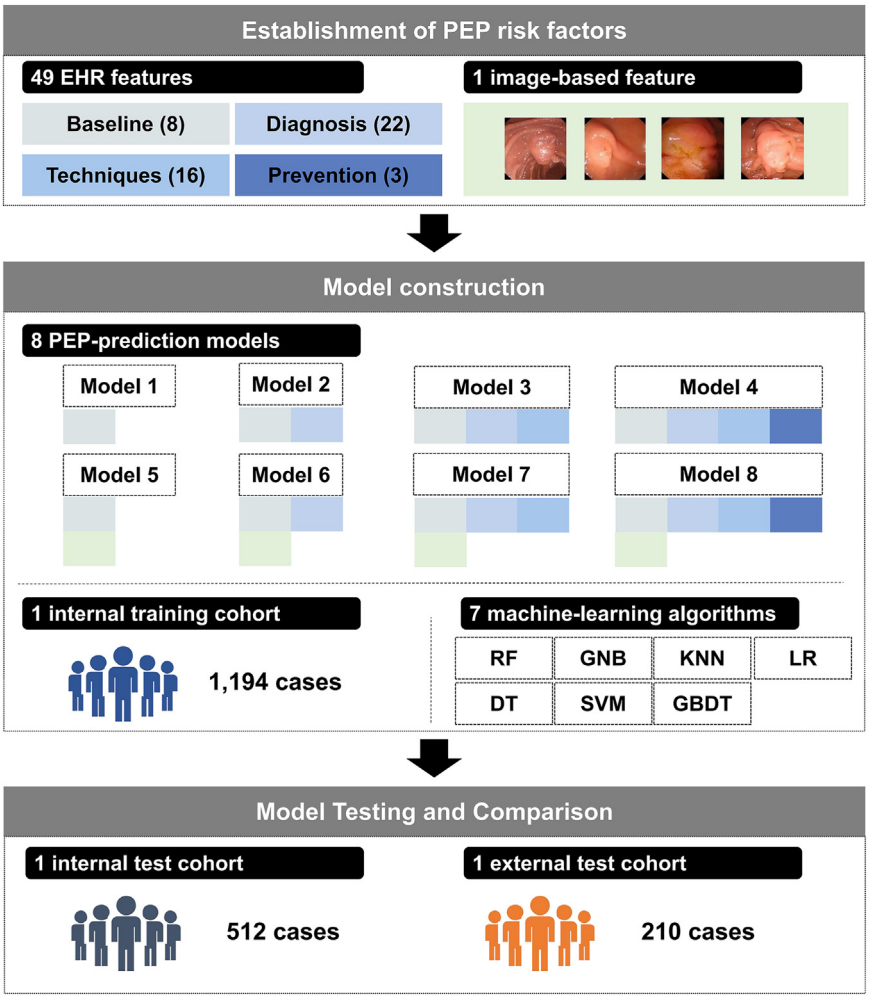


Figure 1. Diagram of the study. Forty-nine EHR features throughout the entire ERCP procedure were divided into 4 categories, and PEP risk factors were established together with the MDP images. For constructing models 1 to 4, 4 feature categories were sequentially incorporated in the training process. Then, the image-based feature was added to multimodal models 5 to 8. Cases (n = 1916) of ERCP were retrospectively collected from multiple centers and used for model construction and validation. *PEP*, Post-ERCP pancreatitis; *EHR*, electronic health record; *RF*, random forest; *GNB*, Gaussian naïve Bayes; *KNN*, k-nearest neighbor; *LR*, logistic regression; *DT*, decision tree; *SVM*, support vector machine; *GBDT*, gradient-boosting decision tree; *MDP*, major duodenal papilla.

presented in Table 1. In the internal test, comparing model 2 with model 1, the addition of diagnosis features significantly improved the accuracy from 61.52% (95% CI, 57.23-65.63) to 74.22% (95% CI, 70.26-77.82; $P < .001$) and significantly improved the specificity from 61.33% (95% CI, 56.68-65.40) to 75.00% (95% CI, 70.92-78.68; $P < .001$). Between model 3 and model 2, the additional input of technique features led to improvement in sensitivity from 63.89% (95% CI, 47.58-77.53) to 77.78% (95% CI, 61.92-88.29; $P = .0975$). When comparing model 4 and model 3, the addition of prevention strategies further improved the accuracy from 75.56% (95% CI, 71.69-79.11) to 85.74% (95% CI, 82.44-88.50; $P < .001$), the sensitivity from 77.78% (95% CI, 61.92-88.29) to 83.33% (95% CI, 64.98-90.25; $P = .0306$), and the specificity significantly from 75.42% (95% CI, 71.36-79.07) to 85.92% (95% CI, 82.51-88.76; $P < .001$).

When adapted from the internal test set to the external test set, EHR-only models showed excellent robustness. Comparing model 3 with model 2, the addition of technique features significantly improved the accuracy from 69.52% (95% CI, 62.99-75.35) to 77.62% (95% CI, 71.52-82.73; $P < .01$), sensitivity from 73.33% (95% CI, 48.05-89.10) to 80.00% (95% CI, 54.81-92.95), and specificity significantly from 69.23% (95% CI, 62.43-75.28) to 77.44% (95% CI, 71.08-82.74; $P < .05$). Between model 3 and model 4, the additional input of prevention strategies led to similar improvements as seen with technique features.

Performance of fusion models

The accuracy of fusion models 5 to 8 in comparing the model enhancement of image-based factor, diagnosis features, technique features, and prevention strategies are presented in Table 1. In the internal test, the overall

TABLE 1. Performance of EHR-only models and fusion models for internal and external validation

Modal groups	Model no.	Features	Internal validation				External validation			
			AUC (95% CI)	Accuracy	Sensitivity	Specificity	AUC (95% CI)	Accuracy	Sensitivity	Specificity
EHR model	1	Baseline	.69 (.60-.78)	61.52 (315/512)	66.67 (24/36)	61.13 (291/476)	.73 (.66-.80)	68.57 (144/210)	73.33 (11/15)	67.18 (131/195)
	2	Baseline + diagnosis	.78 (.73-.83)	74.22 (380/512)*	63.89 (23/36)	75.00 (357/476)*	.75 (.69-.81)	69.52 (146/210)	73.33 (11/15)	69.23 (135/195)
	3	Baseline + diagnosis + techniques	.82 (.75-.90)	75.56 (387/512)	77.78 (28/36)	75.42 (359/476)	.81 (.76-0.86)	77.62 (163/210)†	80.00 (11/15)	77.44 (151/195)‡
	4	Baseline + diagnosis + techniques + prevention strategies	.88 (.82-.94)	85.74 (439/512)*	83.33 (30/36)	85.92 (409/476)*	.90 (.87-.93)	89.52 (188/210)*	86.67 (11/15)	89.74 (175/195)†
Fusion model	5	Image-based factor + baseline	.82 (.77-.87)	84.77 (434/512)§	72.22 (26/36)	85.71 (408/476)§	.85 (.81-.89)	77.62 (163/210)‖	80.00 (11/15)	77.44 (151/195)‖
	6	Image-based factor + baseline + diagnosis	.88 (.84-.92)	86.91 (445/512)§	75.00 (27/36)	87.82 (418/476)§	.89 (.85-.93)	83.81 (176/210)§	86.67 (11/15)	83.59 (163/195)¶
	7	Image-based factor + baseline + diagnosis + techniques	.91 (.86-.96)	90.04 (461/512)§	83.33 (30/36)	90.55 (431/476)§	.93 (.90-.96)	90.95 (191/210)‡,§	86.67 (13/15)	91.28 (178/195)‡,§
	8	Image-based factor + baseline + diagnosis + techniques + prevention strategies	.96 (.94-.98)	92.77 (475/512)§	86.11 (31/36)	93.28 (444/476)§	.98 (.96-1.00)	93.81 (197/210)	100.00 (15/15)	93.33 (182/195)

Values for Accuracy, Sensitivity, and Specificity columns are % (n/N). The McNemar test was used to compare the accuracy, sensitivity, and specificity.
EHR, Electronic health record; *AUC*, area under the receiver-operating characteristic curve; *CI*, confidence interval.
Significant difference between the upper and lower models in the same modal groups: *$P < .001$, †$P < .01$, ‡$P < .05$.
Significant difference between the EHR model and the corresponding fusion model. Models 1-4 correspond to Models 5-8 1 by 1, such as model 1 corresponds to model 5: §$P < .001$, ‖$P < .05$, ¶$P < .01$.

performance of fusion models was significantly higher than that of EHR-only models. Comparing model 8 with model 4, the addition of image-based factors improved the sensitivity from 83.33% (95% CI, 68.11-92.13) to 86.11% (95% CI, 71.34-93.92), significantly improved the accuracy from 85.74% (95% CI, 82.44-88.50) to 92.77% (95% CI, 90.19-94.71; $P < .001$), and significantly improved the specificity from 85.92% (95% CI, 82.51-88.76) to 93.28% (95% CI, 90.67-95.20; $P < .001$).

In the external test, models 5 to 8 demonstrated equivalent levels of performance, indicating the robustness of the fusion models. Between model 8 and model 4, the additional input with image-based factors improved the accuracy from 89.52% (95% CI, 84.64-92.98) to 93.81% (95% CI, 89.70-96.35), sensitivity from 86.67% (95% CI, 62.12-96.27) to 100% (95% CI, 79.61-100), and the specificity from 89.74% (95% CI, 84.69-93.26) to 93.33% (95% CI, 88.93-96.06).

Similar to the EHR-only model, the models showed a similar tendency on the weight ranking of features. The performance of multistep validation for EHR models and fusion models is presented in Figure 2.

Feature contribution analysis

We analyzed the weight and ranking of features, thus identifying the contributions of each feature on the final diagnosis. The contributions of each feature for models 1 to 8 can be found in Supplementary Figures 3-10 (available online at www.giejournal.org).

For the EHR-only models, previous pancreatitis, sphincter of Oddi disorders (SODs), and pancreatic duct stricture were the top 3 features in model 2 (baseline + diagnosis) and still played an important role with the addition of the remaining features. After adding the technique features, the top 5 features in model 3 (baseline + diagnosis + technique) were difficult cannulation, previous pancreatitis, SODs, and needle-knife sphincterotomy (NKS). With the addition of prevention strategies, the top 5 features in model 4 (baseline + diagnosis + technique + prevention strategies) were difficult cannulation, use of NSAIDs, previous pancreatitis, SODs, and NKS. All 3 prevention strategies played an important role in model 4.

For the fusion model 8, the top 5 features were the image-based factor, difficult cannulation, use of NSAIDs, previous pancreatitis, and SODs. The weight of the fusion model and EHR-only model exhibited similar ranking. It is worth mentioning that the image-based factor had the highest weight in all fusion models.

Performance of different machine learning algorithms

Model 8 showed the best performance, and then we compared the precision of different machine learning algorithms within the 50 features (Fig. 3). The random forest algorithm demonstrated superior performance, as shown in Table 2. Additionally, the machine learning model achieved superior performance (model 8, AUC = .96) over traditional logistic regression models (AUC = .87).

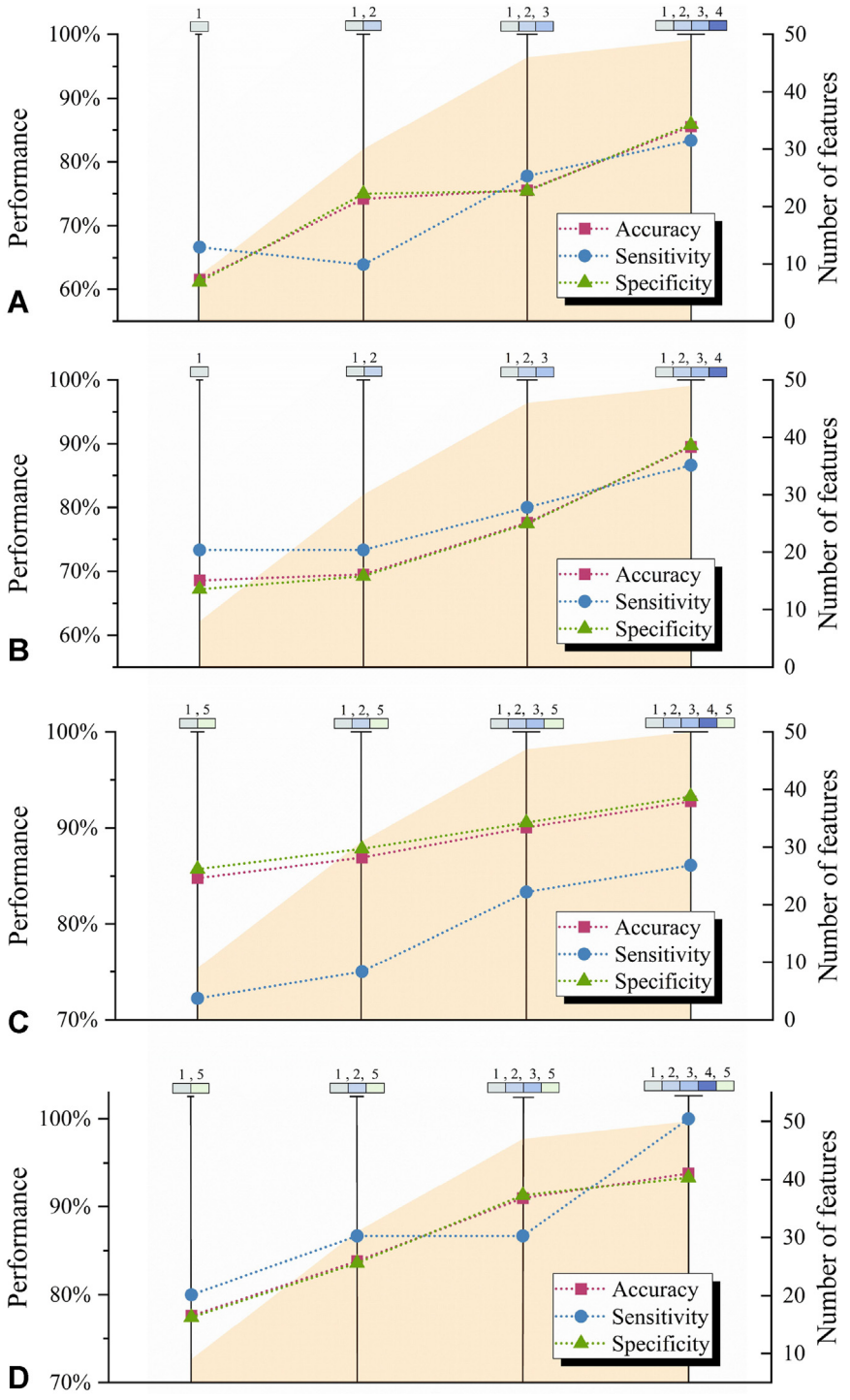


Figure 2. A multistep validation approach using incremental categories on training sets. The cumulative features scheme started with 8 baseline features, followed by the addition of 22 diagnosis features, 16 technique features, 3 prevention strategies, and an image-based factor. With the addition of different features, the performance of the model increased continuously and finally reached an accuracy of 92.77%. **A,** Internal validation of electronic health record (EHR)-only models 1 to 4. **B,** External validation of EHR-only models 1 to 4. **C,** Internal validation of fusion models 5 to 8. **D,** External validation of fusion models 5 to 8.

DISCUSSION

Compared with previous studies, we have taken a novel approach to construct and evaluate 8 models by incrementally incorporating different categories of risk factors and conducted a comprehensive exploration of the contribution of various risk factors. In this study, we established models using multimodal data, which achieved clinical

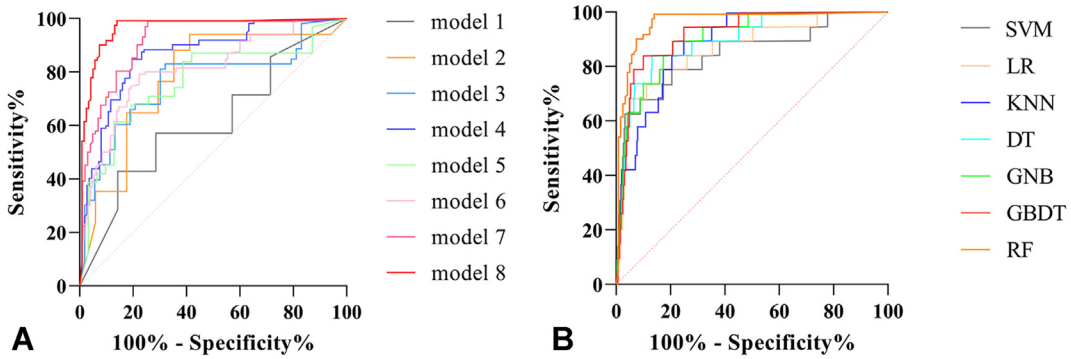


Figure 3. Performance comparison of models 1 through 8 and different machine learning algorithms. **A,** Performance comparison of models 1 through 8. With the addition of baseline features, diagnosis features, techniques features, prevention strategies, and an image-based factor, the performance of the model increased continuously and finally reached an area under the receiver-operating characteristic curve (AUC) of .96. **B,** Performance comparison of different machine learning algorithms. Seven different machine learning algorithms were compared within the 50 features. The random forest algorithm shown the best performance and achieved superior performance over traditional logistic regression models (AUC = .87). *RF*, Random forest; *GNB*, Gaussian naïve Bayes; *KNN*, k-nearest neighbor; *LR*, logistic regression; *DT*, decision tree; *SVM*, support vector machine; *GBDT*, gradient-boosting decision tree.

TABLE 2. Performance of various machine learning algorithms was evaluated using all 50 features

Algorithm	Area under the receiver-operating characteristic curve (95% confidence interval)	Accuracy	Sensitivity	Specificity
Random forest	.96 (.94-.98)	84.57 (433/512)	77.78 (28/36)	85.08 (405/476)
Gradient-boosting decision tree	.92 (.90-.94)	84.96 (435/512)	77.78 (28/36)	85.50 (407/476)
Gaussian naïve Bayes	.90 (.88-.92)	86.13 (441/512)	80.56 (29/36)	86.55 (412/476)
Decision tree	.89 (.86-.92)	87.30 (447/512)	83.33 (30/36)	87.61 (417/476)
k-Nearest neighbor	.89 (.85-.91)	88.48 (453/512)	83.33 (30/36)	88.87 (423/476)
Logistic regression	.87 (.84-.90)	90.04 (461/512)	86.11 (31/36)	90.34 (430/476)
Support vector machine	.86 (.82-.89)	92.77 (475/512)	86.11 (31/36)	93.28 (444/476)

Values for Accuracy, Sensitivity, and Specificity columns are % (n/N).

acceptable performance to predict high-risk patients with PEP. The full-feature prediction model 8 demonstrated the highest performance among all models, highlighting the significance of incorporating image information and technique features during the ERCP procedure for effective risk stratification of PEP. Notably, the preprocedure decision-making model 6 was the best model for identifying patients who will be at high risk for PEP, achieving an accuracy of 86.91% and an AUC of .88.

According to a recent report by the Healthcare Cost and Utilization Project, the economic impact of PEP is substantial.[2] Additionally, PEP is the leading cause of lawsuits associated with ERCP.[28] Thus, the American Society for Gastrointestinal Endoscopy guidelines emphasize the importance of accurately determining the personalized risk of PEP, because it can lead to a more precise risk assessment and enhance decision-making regarding interventions.[1] The ability to identify patients who are likely to develop PEP could aid in appropriately selecting individuals who can potentially be kept for observation after the procedure and/or get more-aggressive preventative measures for prevention management of PEP. On the other hand, for low-risk individuals, avoiding preventive strategies such as pancreatic duct stents and rectal NSAIDs can not only reduce the treatment burden on patients but also significantly decrease the costs.

Prior research has indicated that the macroscopic shape of the MDP can aid in identifying patients with an elevated risk of PEP, which is associated with difficult cannulation.[27,29-33] In this study, we combined MDP images with EHR features to rank the risk of PEP across different phases of ERCP. This approach allows for the selection of personalized prevention strategies based on individual risk profiles. Furthermore, our analyses demonstrate that MDP images offer valuable information for predicting PEP and bile duct cannulation. This information can assist endoscopists in choosing the most appropriate cannulation technique and reducing the risk of failed attempts, which can lead to irreversible adverse events associated with PEP.

Our data supported the association between established techniques of ERCP and the development of PEP.

Specifically, difficult cannulation, NKS, and endoscopic sphincterotomy played significant roles in predicting PEP within our comprehensive model. Based on current clinical evidence, if cannulation attempts have already failed, alternative techniques such as NKS can be considered to prevent further escalation in the risk of PEP.[6,34,35] However, with the acquisition of MDP images, our model 6 enabled an early evaluation of the risk of PEP, allowing for the selection of alternative cannulation techniques to avoid multiple unsuccessful attempts, which can significantly increase the risk of PEP development. Overall, our model provided a more precise and individualized approach to accurately identify high-risk patients and promptly intervene in their care.

A previous study indicated that certain high risk factors, such as female sex, young age, and a history of previous pancreatitis, are associated with PEP.[1] However, as demonstrated by model 2, which included baseline and diagnosis information, the predictive performance was poor. This suggests that patient-related factors alone may be helpful but not sufficient for accurate PEP prediction. In our study, because models 1 and 2 were basic model training for baseline and diagnosis features, we fitted 2 further models with the addition of endoscopist-driven features including diagnosis and techniques. The performance of model 4 is significantly better than that of model 2, suggesting that effective risk control by experienced endoscopists plays a crucial role in preventing PEP. Therefore, we speculated that the prevention of PEP heavily relies on implementing active preventive strategies for high-risk groups, selecting appropriate techniques, and minimizing contact with the pancreatic duct during the procedure.

This study offers several advantages over previous studies (Supplementary Table 4, available online at www.giejournal.org).[20-26] First, we carefully selected and categorized 49 EHR features based on the different phases of ERCP, including preprocedure, intraprocedure, and periprocedure phases, aligning with the real clinical workflow. Notably, we incorporated endoscopic diagnosis, technique features, and prevention strategies to address specific clinical challenges, enhancing the adaptability to different clinical needs. Second, compared with the traditional logistic regression method, which was widely used previously, other machine learning approaches provided significantly better performance. We speculated that the prediction of PEP was a nonlinear and complex issue that probably interacts with many variables. Our multimodal prediction models were constructed based on a nonlinear method, which could help to unravel questions from traditional linear models. Third, our prediction model exhibited robust predictive performance when tested against an independent multicenter external dataset, validating its generalizability beyond the training dataset. This suggests that our prediction model has the potential to achieve favorable performance across diverse external datasets, although further validation is necessary to substantiate this claim.

Nevertheless, this study has several limitations that should be acknowledged. First, we performed a retrospective study. Prospective, large-scale, multicenter, validation studies are needed to obtain high-level evidence for clinical use in future research. Second, independently developing a severity assessment model to distinguish between mild to severe pancreatitis is helpful for clinical application. One potential hypothesis is that the predictive factors for mild pancreatitis and severe pancreatitis may differ. Therefore, further work could focus on severity assessment of PEP in large multicenter cohorts. Third, as ERCP techniques continue to evolve and become more complex, the features included in this study may need to be updated. Furthermore, exploring the use of a large language model to automatically extract multimodal features from ERCP records could be of great interest.

In summary, this study has successfully developed prediction models with multistep validation, incorporating various categories of EHR data and image-based factors, to assess the risk of developing PEP in ERCP patients. These results establish a solid foundation for future exploration of artificial intelligence technologies in this specific patient population.

DISCLOSURE

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Individual deidentified participant data that underlie the results reported in this article can be shared with investigators for research purposes. To gain access, data requesters will need to contact the corresponding author.

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Abbreviations: AUC, area under the receiver-operating characteristic curve; CI, confidence interval; EHR, electronic health record; ICD-11, International Classification of Diseases, 11th revision; MDP, major duodenal papilla; PEP, post-ERCP pancreatitis; NKS, needle-knife sphincterotomy; NSAID, nonsteroidal anti-inflammatory drug; SOD, sphincter of Oddi disorders.

*Drs Xu and Dong contributed equally to this article.

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Current affiliations: Key Laboratory of Hubei Province for Digestive System Disease (1), Hubei Provincial Clinical Research Center for Digestive Disease Minimally Invasive Incision (2), Renmin Hospital of Wuhan University, Wuhan, China; Department of Gastroenterology, Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China (3).

Reprint requests: Honggang Yu, MD, PhD, Department of Gastroenterology, Renmin Hospital of Wuhan University, 99 Zhangzhidong Rd, Wuhan 430060, Hubei Province, China.

APPENDIX 1

Architecture of the prediction model

Pretrained ResNet-50 deep neural network models were used, with a batch size of 32, an initial learning rate of .001, Nesterov momentum of .90, and categorical cross-entropy loss. In addition, image augmentation procedures were used, as follows, to increase the dataset size and strengthen model generalizability: rotation (clockwise by 0 to 180 degrees, selected randomly), horizontal flip, and vertical flip. Brightness and scale adjustment were also performed for training inputs. Local contrast normalization of the major duodenal papilla (MDP) images took place after the central disc of the photo had been extracted to a template image of 512 × 512 pixels. These networks were trained with 2 output nodes corresponding to the 2 target classes for low post-ERCP pancreatitis (PEP) risk and high PEP risk. Training was performed until convergence on accuracy was demonstrated with the internal validation data. An early-stop procedure was applied to avoid overfitting: The training was stopped if the loss on the validation set no longer decreased for 5 epochs.

Development of the electronic health record (EHR) data–only models involved extraction of the relevant clinical feature scores for each patient. Subsequently, a grid search with a 5-fold cross-validation was used to determine the optimal random forest hyperparameter values for each model. The hyperparameters were optimized for the number of estimators, maximum number of features, maximum depth, and split quality criterion. Once the optimal hyperparameter values were determined, the actual random forest model was trained on the entire training data.

Preliminary experiments have been attempted using feature vectors extracted just before the output node layer, of 10 and 100 nodes, respectively. The final random forest model using just the output node values generally outperformed models using a larger number of features extracted from previous intermediate layers. As such, the output node layer values were used to develop the actual experimental models as described below.

From the trained image-only models, image-based scores were generated for each patient. All images for a case were first processed by the model to produce an image-level score. These image-level scores were calculated from the individual output node value by multiplying the value of each node with the index of the node (which corresponds to the severity of the condition) and then summing these values into a single representative score for that condition.

The process of developing EHR data–only models encompassed the extraction of pertinent clinical feature scores for each individual case. After this, a grid search technique was used, along with 5-fold cross-validation, to identify the most suitable hyperparameter values for each model in the machine learning algorithms. The hyperparameters that were optimized included the number of estimators, maximum number of features, maximum depth, and split quality criterion. Once the optimal hyperparameter values were determined, the machine learning algorithms were trained using the complete training dataset.

Image-only model

The image-only model was trained for extracting PEP-related features and predicting the PEP probability using 367 positive and 5007 negative MDP images in dataset 1. Before training, all MDP images were initially transformed into a square template to isolate the central chroma region, followed by contrast normalization. Then, the image-only model underwent training using an independent deep convolutional neural network on the training set to extract image features. The images were not partitioned into distinct sets for training, validation, and testing purposes. The training process involved using k-fold (k = 10) cross-validation to evaluate the performance of the model in predicting PEP. The aforementioned model was designed for binary-category classification, specifically targeting the determination of the image PEP risk score.

This involved using preprocessed MDP images and pretrained ResNet-50 deep neural network models. Subsequently, the aforementioned image models were used to generate a score based solely on the images for each case. To conduct internal validation, dataset 2 comprised 185 MDP images from 36 PEP cases and 1913 MDP images from 476 non-PEP cases. For external validation, the dataset 3 consisted of 52 images from 15 PEP cases and 616 MDP images from 195 non-PEP cases. The image-only model was used to generate a score for each individual image, and subsequently, the case score was calculated by averaging all image scores within the respective case. The image-only model was not required to provide a judgment regarding the presence or not of PEP.

Definition of the primary endpoints

Difficult cannulation. In this study, difficult selective biliary cannulation was defined referring to the European Society of Gastrointestinal Endoscopy criterion: after 5 minutes, 5 failed attempts, or 1 unintended pancreatic duct cannulation.[1]

Hyperamylasemia. Hyperamylasemia was defined as an increase in serum amylase level of at least 3 times greater than the normal upper limit at 24 hours after the procedure, irrespective of whether symptoms were present.[2]

Post-ERCP pancreatitis. PEP was diagnosed when new-onset or worsened abdominal pain lasted for more than 24 hours and was associated with an increase in serum amylase level of at least 3 times greater than the normal upper limit at 24 hours after the procedure.[2,3]

Suspected sphincter of Oddi dysfunction. In this study, "suspected sphincter of Oddi dysfunction" referred to the pre-ERCP suspicion of a functional or structural abnormality of the sphincter of Oddi, independent of any manometric findings, believed to be the cause of recurrent abdominal pain or pancreatitis.[4]

Cannulation time. Cannulation time was measured between the time when the cannula or papillotome was advanced out of the endoscope channel, in front of the papilla, and the time when successful deep cannulation was evident by injection of contrast or the time when the procedure was abandoned because of unsatisfactory cannulation.

Text definition of baseline features

- *Age.* Patients aged <60 years were labeled positive and those aged ≥60 years were labeled negative.
- *Sex.* Female patients were labeled positive and male patients were labeled negative.
- *Body mass index.* Patients with body mass index <25 kg/m^2 were labeled positive and those with body mass index ≥25 kg/m^2 were labeled negative.
- *Previous acute pancreatitis.* In this study, difficult selective biliary cannulation was defined referring to the European Society of Gastrointestinal Endoscopy criterion: after 5 minutes, 5 failed attempts, or 1 unintended pancreatic duct cannulation.[1]
- *Active smoking.* On admission, the patient's medical records noted a history of smoking habit.
- *Surgically altered anatomy.* Patients with a surgical history, which includes alterations to the anatomic structures relevant to duodenum, such as Billroth surgery or Roux-en-Y gastric bypass surgery.
- *Dilated bile duct (6 mm).* Patients with dilated bile duct ≥6 mm were labeled positive and those with <6 mm were labeled negative.
- *Normal bilirubin level (1.2 mg/dL).* Patients with normal bilirubin <1.2 mg/dL were labeled positive and those with ≥1.2 mg/dL were labeled negative.

Text definition of diagnosis features

- *Small papilla.* Small, often flat, visually with a diameter not bigger than 3 mm (approximately 2 sphincterotome diameter).
- *Fistula of gallbladder or bile duct.* Biliary fistula is a rare adverse event of gallstones, defined as chronic tubular ulcers.
- *Bile duct sludge.* Biliary sludge is a suspension of cholesterol monohydrate crystals or calcium bilirubinate granules found predominantly in the gallbladder. In this study, biliary sludge was defined as amorphous tiny bile duct stones observed in fluoroscopic images.
- *Normal bile duct stone.* Different from bile duct sludge, multiple bile duct stones and bile duct–impacted stones, with no clear characteristics.
- *Multiple bile duct stones.* Define as ≥2 stones observed in fluoroscopic images at the same time.
- *Huge bile duct stone (>10 mm).* Bile duct stones larger than 10 mm in diameter.
- *Impacted bile duct stones.* Impacted stones refer to bile duct stones could not be removed through the sphincterotomy and are resistant to fragmentation.
- *Intrahepatic cholelithiasis.* Intrahepatic stones are part of the calculus of the bile duct, referring to the stones located proximal to the confluence of the left and/or right hepatic ducts.
- *Calculus of gallbladder or cystic duct.* The stone is in the gallbladder or cystic duct.
- *Mirizzi syndrome.* Endoscopic diagnosis of Mirizzi syndrome. The common hepatic duct demonstrates extrinsic narrowing, accompanied by dilation of the intrahepatic bile ducts, with the gallbladder not visualized. Exerted pressure on the common hepatic duct often manifests as a rightward concavity, representing a characteristic orientation of the cystic duct relative to the common hepatic duct.
- *Biliary stricture.* Biliary strictures or bile duct strictures refer to segments of narrowing of the intrahepatic or extrahepatic biliary ductal system. In the study, a positive label was defined as the presence of a narrowed bile duct in fluoroscopic images or diagnosis in the ERCP report.
- *Pancreatic duct stricture.* Pancreatic duct stricture refers to the narrowing or obstruction of the pancreatic duct, the conduit for pancreatic excretions. In this study, a positive label was defined as the presence of a narrowed pancreatic duct in fluoroscopic images or diagnosis in the ERCP report.
- *Dilated pancreatic duct.* Pancreatic duct dilation refers to the enlargement of the main pancreatic duct or its branches within the pancreas. In this study, a positive label was defined as the presence of a dilated pancreatic duct in fluoroscopic images or diagnosis in the ERCP report.
- *Occupied lesion (cystic diseases or neoplasms of pancreas).* In this study, occupied lesion was summarized as cystic disease or neoplasms of pancreas.
- *Intraoperative bleeding.* ERCP-associated bleeding refers to bleeding that occurs during the ERCP procedure and is 1 adverse event associated with ERCP.
- *Cholangitis.* In this study, a positive label for cholangitis was defined as the presence of diagnosis in the ERCP report.
- *Cholecystitis.* In this study, a positive label for cholecystitis was defined as the presence of diagnosis in the ERCP report.
- *Diverticulum of the duodenum.* In this study, a positive label for diverticulum of the duodenum was defined as

the presence in duodenoscopy images or diagnosis in the ERCP report.
- *Choledochopancreatic junction anomaly.* In this study, a positive label for a choledochopancreatic junction anomaly was defined as the presence in fluoroscopic images or diagnosis in the ERCP report.
- *Sphincter of Oddi disorders.* In this study, "suspected sphincter of Oddi dysfunction" referred to the pre-ERCP suspicion of a functional or structural abnormality of the sphincter of Oddi, independent of any manometric findings, believed to be the cause of recurrent abdominal pain or pancreatitis.
- *Duodenal ulcer.* In this study, a positive label for duodenal ulcer was defined as the presence in duodenoscopy images or diagnosis in the ERCP report.
- *Specified acquired anatomic alterations of gallbladder or bile duct.* Patients with a surgical history that included alterations to the anatomic structures relevant to the gallbladder or bile duct, such as postcholecystectomy. The patient's fluoroscopic images usually show postcholecystectomy changes.

Text definition of technique features

- *Therapeutic ERCP.* The therapeutic intent of ERCP was outlined in the pre-ERCP checklist.
- *Experience of the operator.* In a previous clinical trial conducted by our research team, we surveyed the experience of all endoscopists performing ERCP at our center. Based on their years of experience and lifetime procedural volume, we categorized their operation levels into 3 grades.
- *Needle-knife sphincterotomy or precut sphincterotomy.* In this study, a positive label was defined as being visible in the ERCP videos and images or being described in the report.
- *Transpancreatic sphincterotomy.* In this study, a positive label was defined as being visible in the ERCP videos and images or being described in the report.
- *Difficult cannulation.* In this study, difficult selective biliary cannulation was defined referring to the European Society of Gastrointestinal Endoscopy criterion: after 5 minutes, 5 failed attempts, or 1 unintended pancreatic duct cannulation.
- *Procedure time > 20 minutes.* We used the time recorded on duodenoscopy videos or images, starting with insertion and ending with withdrawal, to calculate the procedural time. In this study, a positive label was defined as over 20 minutes.
- *Endoscopic Papillary Balloon Dilatation (≤10 mm).* EPBD (≤10 mm) is visible in the ERCP videos and images or described in the report.
- *Endoscopic Papillary Large Balloon Dilatation (>12 mm).* EPLBD (>12 mm) is visible in the ERCP videos and images or described in the report.
- *Bile duct stent.* Bile duct stent is visible in the ERCP videos and images or described in the report.
- *Mechanical lithotripsy.* Mechanical lithotripsy is visible in the ERCP videos and images or described in the report.
- *Endoscopic sphincterotomy.* Endoscopic sphincterotomy is visible in the ERCP videos and images or described in the report.
- *Endoscopic retrograde biliary drainage.* Endoscopic retrograde biliary drainage is visible in the ERCP videos and images or described in the report.
- *Endoscopic nasobiliary drainage.* Endoscopic nasobiliary drainage is visible in the ERCP videos and images or described in the report.
- *Endoscopic nasopancreatic drainage.* Endoscopic nasopancreatic drainage is visible in the ERCP videos and images or described in the report.
- *Biopsy sampling.* Forceps biopsy sampling or brush cytology is visible in the ERCP videos and images or described in the report.
- *Indwelling T-tube.* Indwelling T-tube is visible in the ERCP videos and images or described in the report.

Text definition of prevention strategy features

- *Pancreatic duct stent placement.* Pancreatic duct stent placement is visible in the ERCP videos and images or described in the report.
- *Rectal and per-procedural fluid solutions.* Use of nonsteroidal anti-inflammatory drugs and fluid solutions was extracted manually from the patient's EHR dataset.

Methods for handling missing data

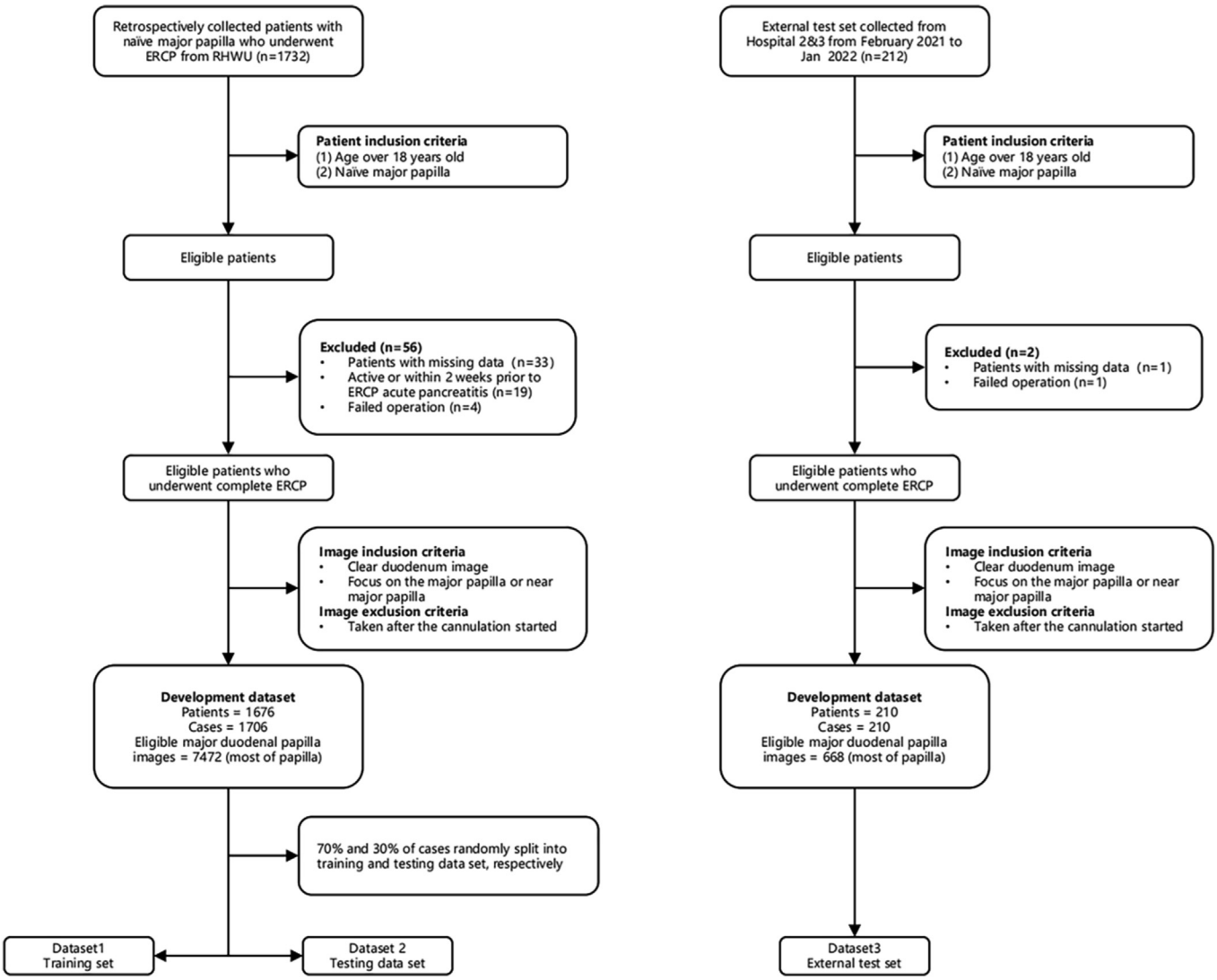
In addressing the issue of missing data, we implemented clear categorized measures to ensure the completeness and accuracy of our data. First, we used a systematic review of the EHR dataset. The EHR database is composed of ERCP reports, duodenoscopy images, ERCP videos, and clinical records. The baseline, diagnosis, and prevention strategy features were obtained from the ERCP reports and clinical records. In most cases, videos supplemented by a comprehensive set of images provided all the necessary technical details. Endoscopists systematically captured images at critical workflow junctures (seeing MDP, the first contact with the papilla of cannulation, endoscopic sphincterotomy electrification and fogging, etc) and these, along with succinct codes in ERCP reports (Endoscopic Retrograde Cholangiograph + endoscopic sphincterotomy + endoscopic nasobiliary drainage), detailed the primary operations performed. In some cases implicitly missing data occurred, for instance, features like small papilla and procedure time were not explicitly documented in the reports, requiring manual extraction or calculation by reviewing ERCP videos. Systematically reviewing the dataset enabled us to extract these features.

Second, we used missing data imputation. We determined the feasibility of imputation based on the reasons for missing data and then selected the appropriate method according to the type of variables. Missing completely at random was defined as data missing because of technologic issues, data entry errors, or occasional problems in sample handling. In this situation, binary classification variables treat missing values as a new category labeled as "NA," whereas continuous and discrete variables use the mean imputation method. For not missing at random, because of the brief ERCP reports or a lack of emphasis on image capture and video recording, it becomes impossible to extract features from the EHR dataset. In this situation, imputation was carefully selected to preserve integrity and validity of the data. After most imputations, we reviewed the extracted dataset. For cases where not missing at random resulted in minor missing data (3.05%, 52/1706), defined as cases with $\leq$3 missing values, the imputation methods were consistent with missing completely at random. Notably, MDP images are excluded from missing data as they are always present for multimodal training.

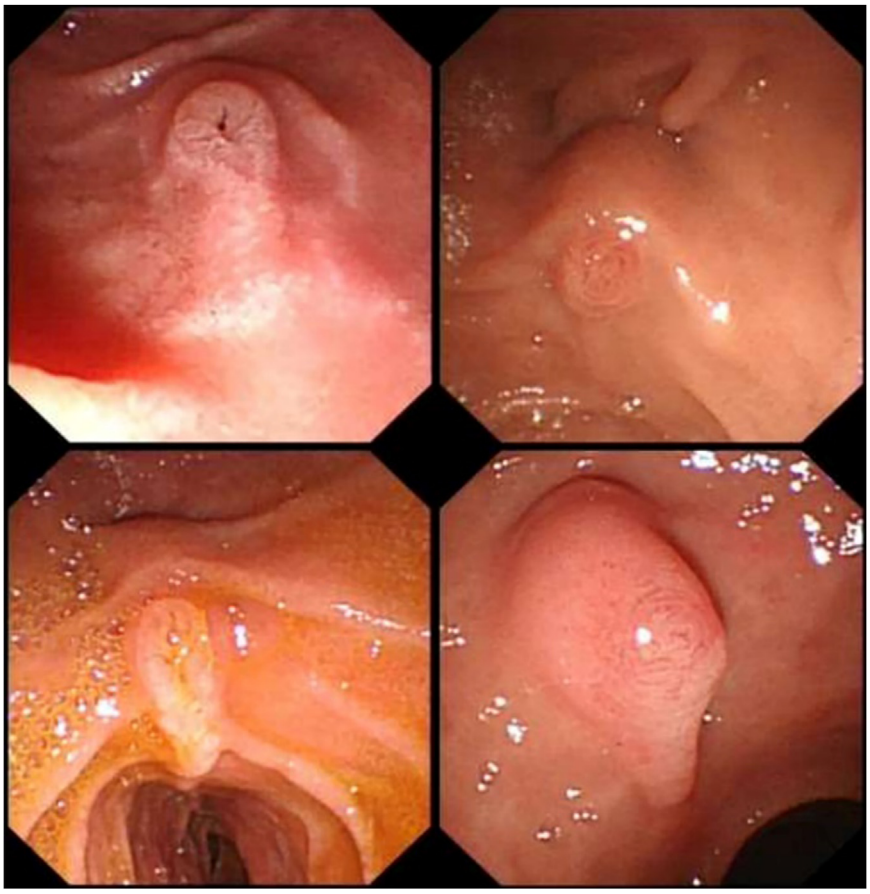
Third, we used complete case analysis for cases where substantial missing data (1.91%, 33/1732) is defined as having $\geq$4 missing values, in which case they were excluded from this study, complying with a complete case analysis strategy.

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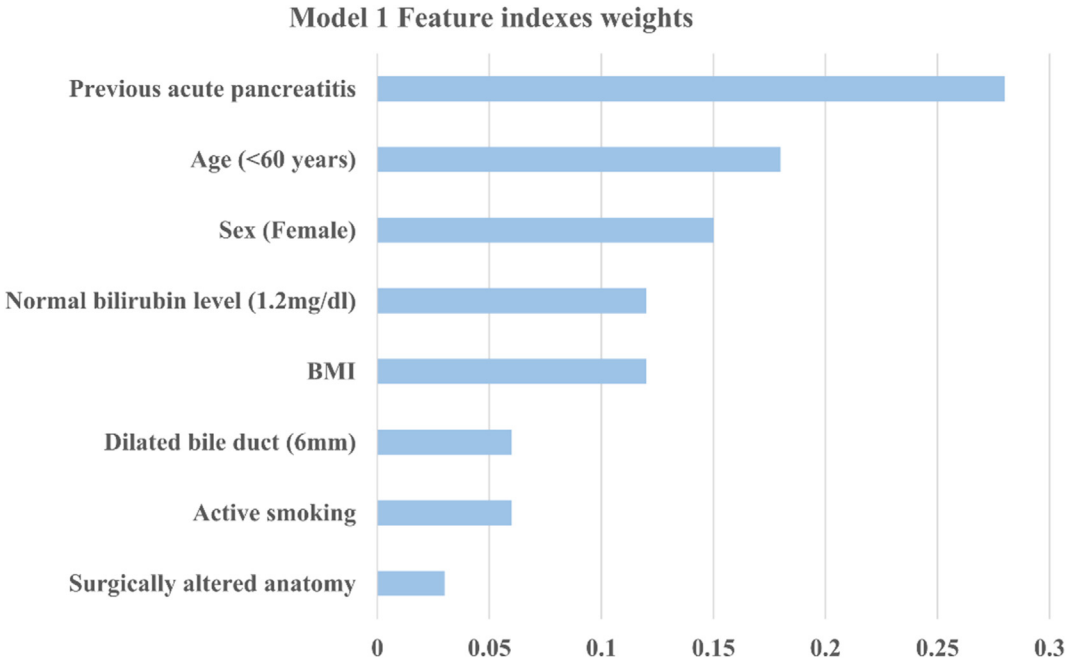
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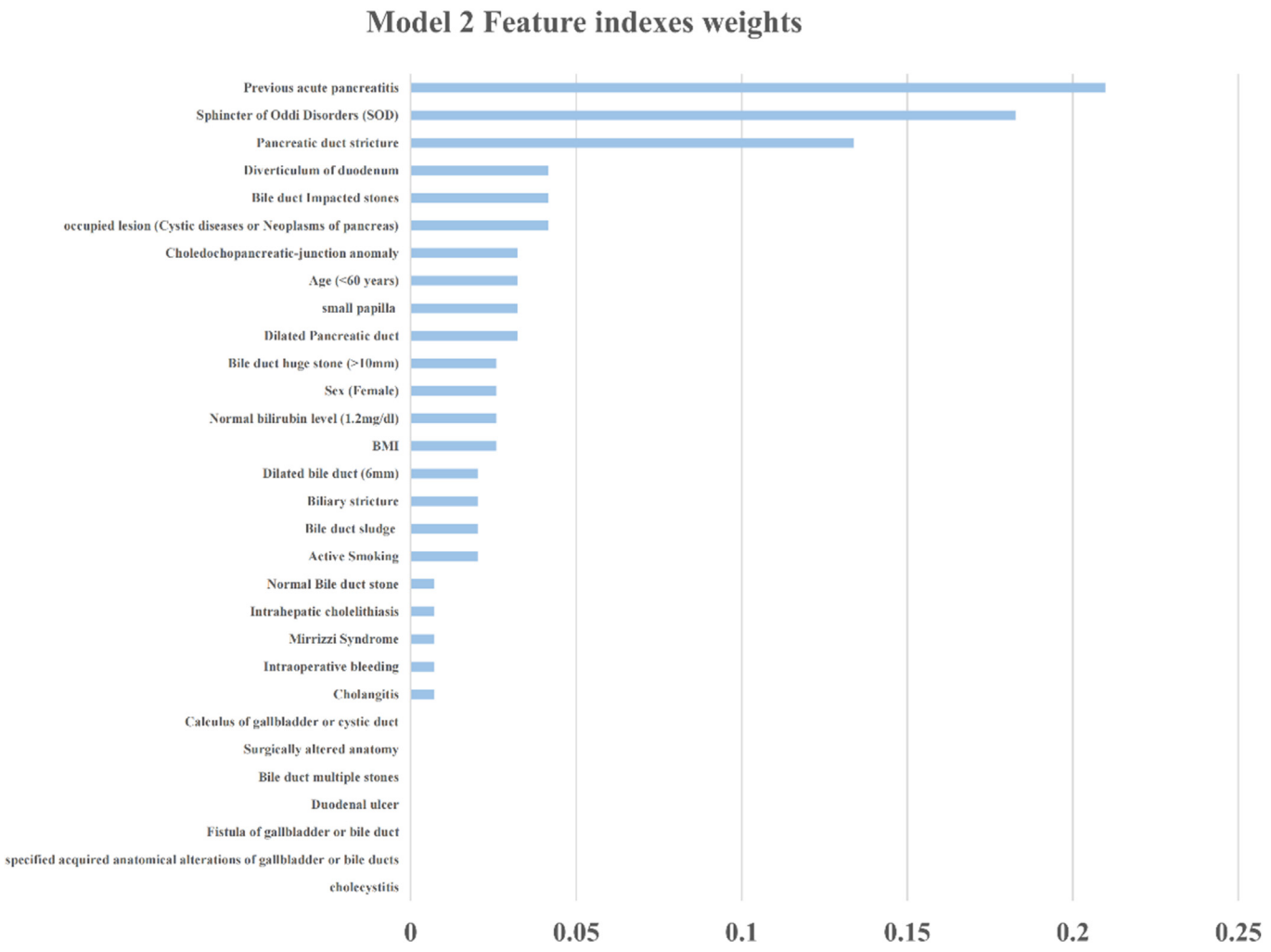
Supplementary Figure 1. Flowchart of the eligibility of the patients and cases in the study. *RHWU*, Renmin Hospital of Wuhan University.



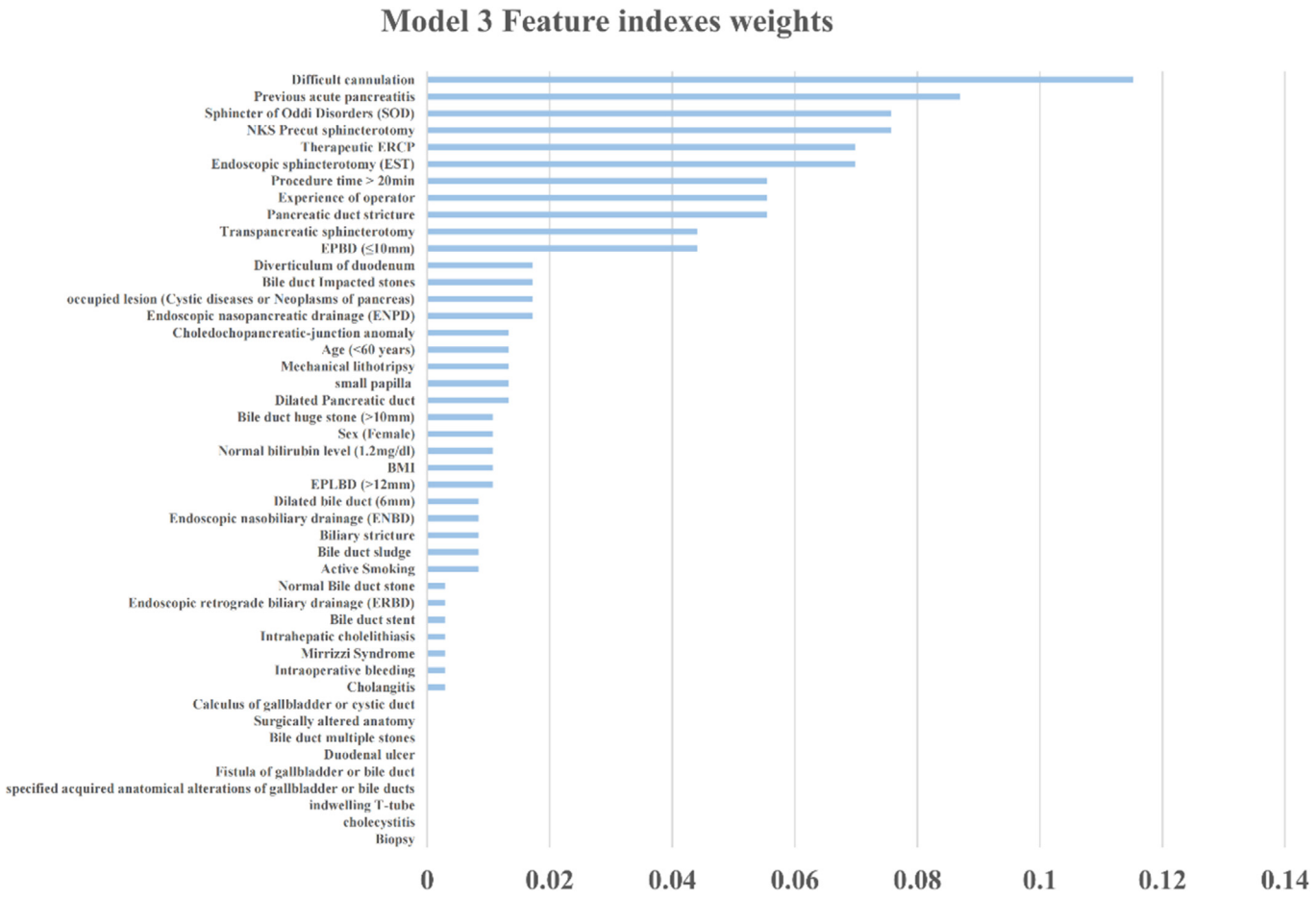
Supplementary Figure 2. Examples of eligible major duodenal papilla images. The deep convolutional neural network generated individualized risk scores based on the images for each case.



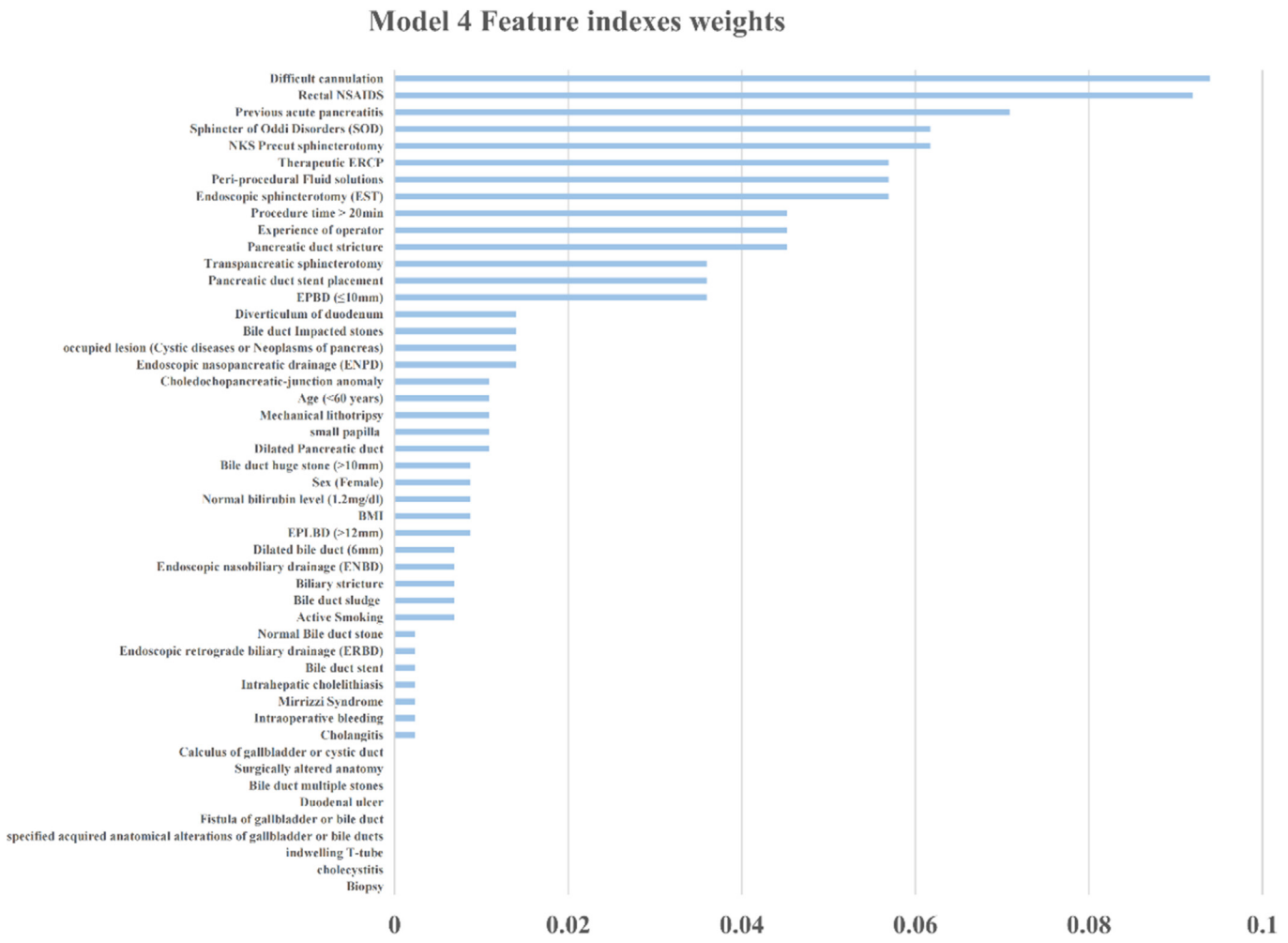
Supplementary Figure 3. The contributions of 8 electronic health record features for model 1. *BMI*, Body mass index.



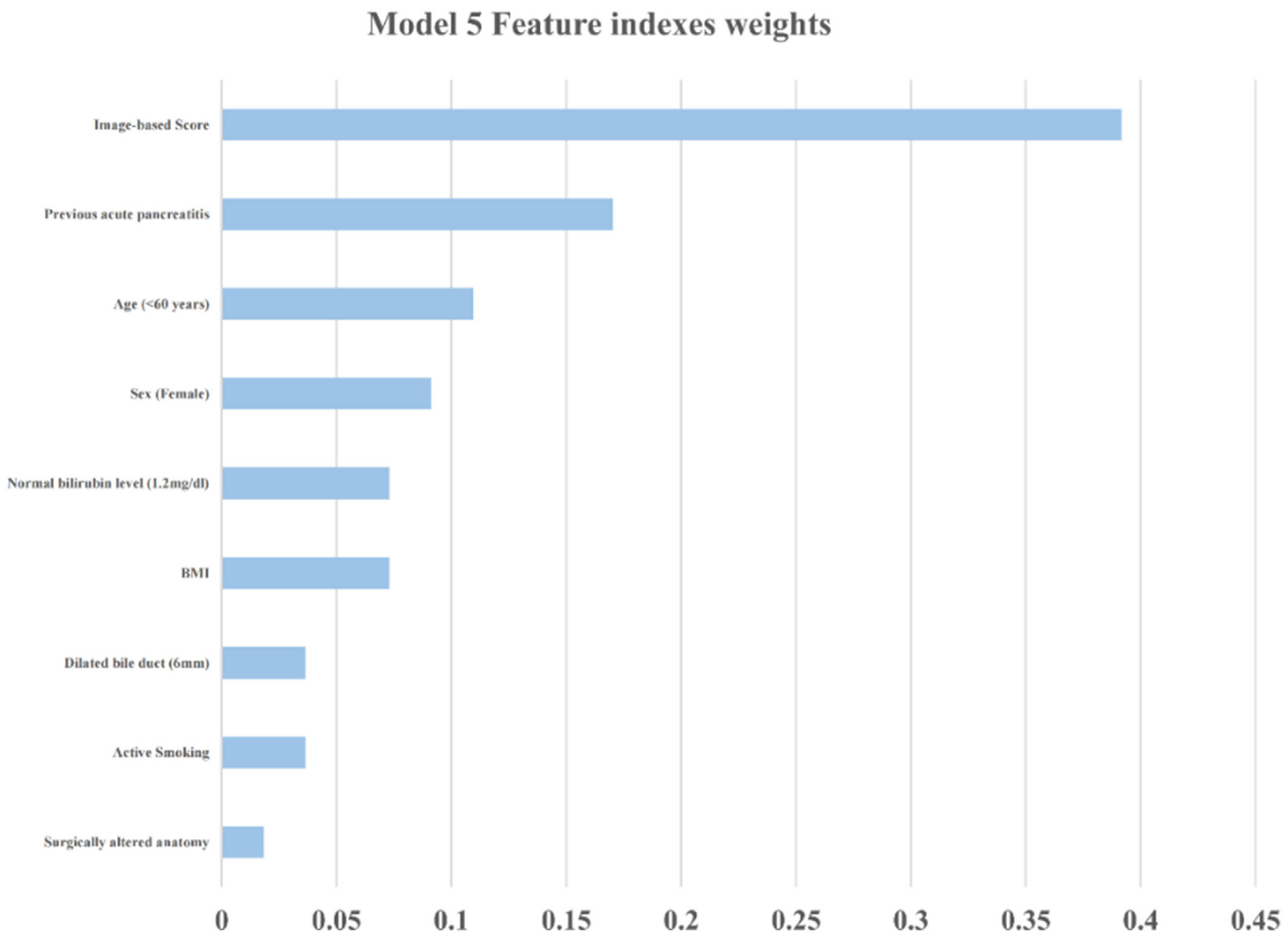
Supplementary Figure 4. The contributions of 30 electronic health record features for model 2. *BMI*, Body mass index.



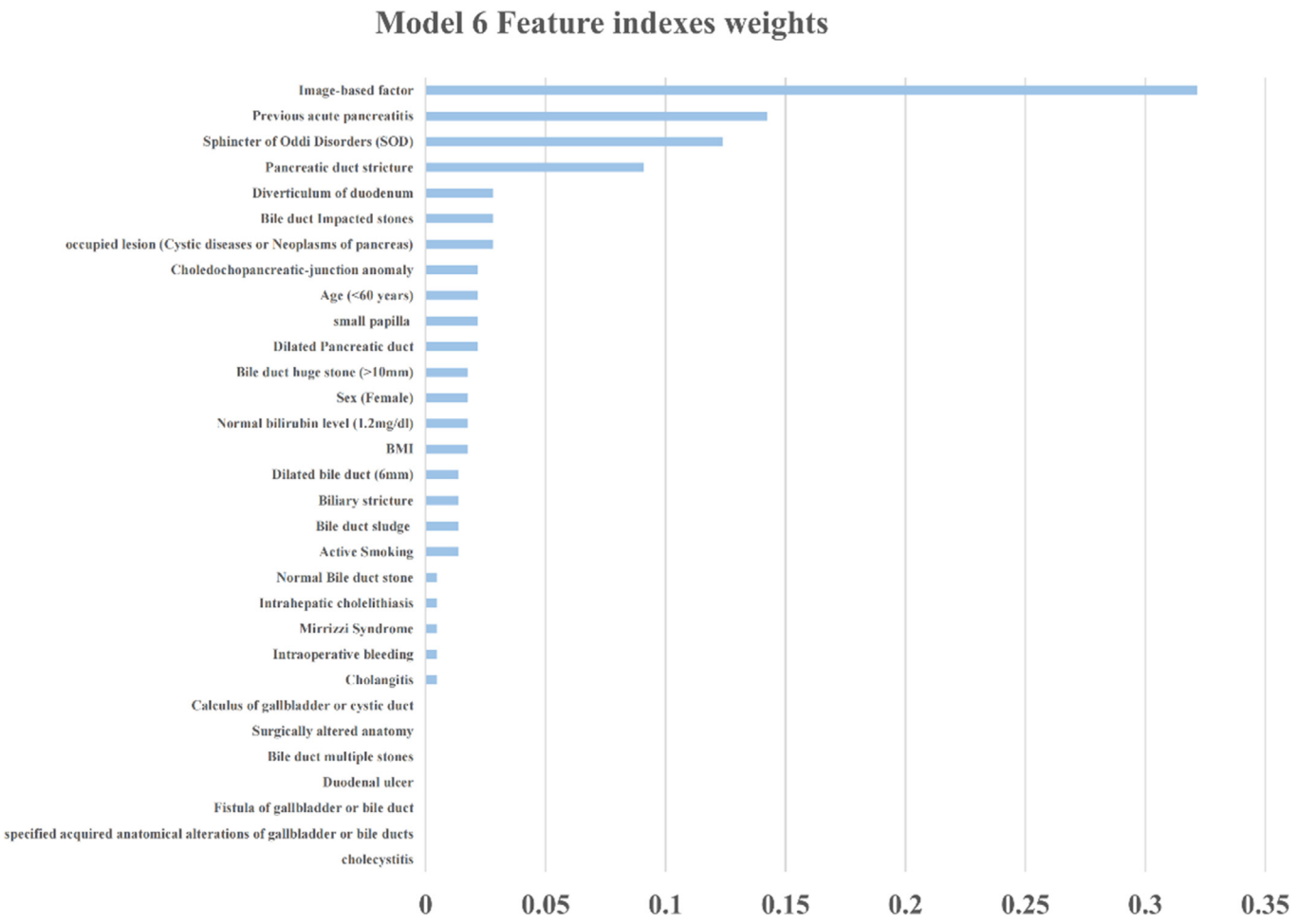
Supplementary Figure 5. The contributions of 46 electronic health record features for model 3. *EPBD*, Endoscopic Papillary Balloon Dilatation; *BMI*, body mass index; *EPLBD*, Endoscopic Papillary Large Balloon Dilatation.



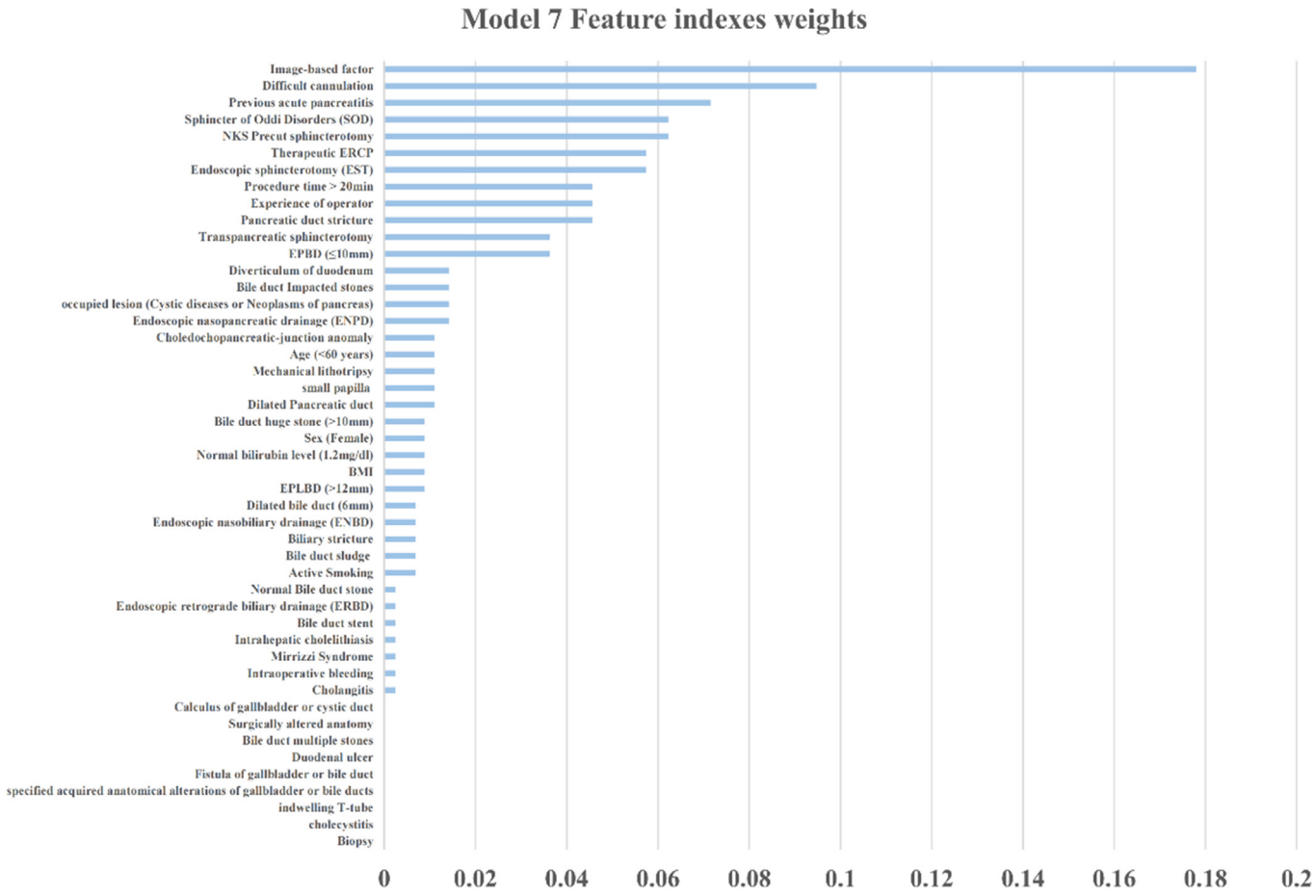
Supplementary Figure 6. The contributions of 49 electronic health record features for model 4. *EPBD*, Endoscopic Papillary Balloon Dilatation; *BMI*, body mass index; *EPLBD*, Endoscopic Papillary Large Balloon Dilatation.



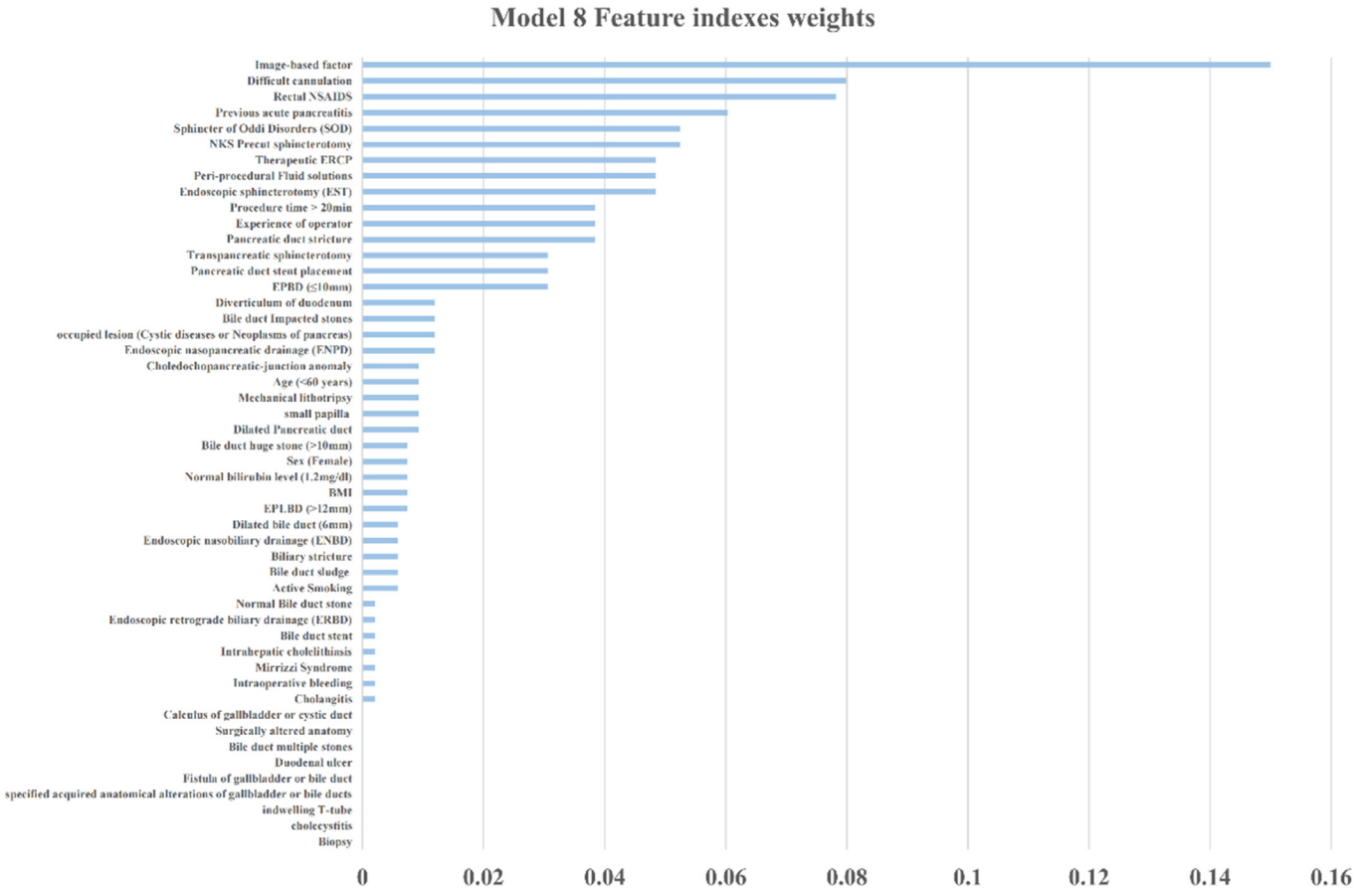
Supplementary Figure 7. The contributions of 8 electronic health record features and image-based factor for model 5. *BMI*, Body mass index.



Supplementary Figure 8. The contributions of 30 electronic health record features and image-based factor for model 6. *BMI*, Body mass index.



Supplementary Figure 9. The contributions of 46 electronic health record features and image-based factor for model 7. *EPBD*, Endoscopic Papillary Balloon Dilatation; *BMI*, body mass index; *EPLBD*, Endoscopic Papillary Large Balloon Dilatation.



Supplementary Figure 10. The contributions of 49 electronic health record features and image-based factor for model 8. *EPBD*, Endoscopic Papillary Balloon Dilatation; *BMI*, body mass index; *EPLBD*, Endoscopic Papillary Large Balloon Dilatation.

SUPPLEMENTARY TABLE 1. Demographic characteristics, post-ERCP pancreatitis rate, and number of images in datasets 1, 2, and 3

	Hospital				Bonferroni-adjusted *P* value	
Demographic characteristics	**Hospital 1**	**Hospital 2**	**Hospital 3**	***P* value**	**Hospital 1 vs 2**	**Hospital 1 vs 3**
Mean age, y (standard deviation)	62.33 (15.10)	64.09 (15.12)	64.32 (14.27)	<.0001	<.0001	<.0001
Female sex	884 (51.82)	33 (50.77)	74 (51.03)	.056	—	—
Post-ERCP pancreatitis rate	118 (6.92)	5 (7.69)	10 (6.90)	.072	—	—

Values are n (%) unless otherwise defined.
—, Not available.

SUPPLEMENTARY TABLE 2. Forty-nine EHR features categorized into baseline characteristics, diagnosis, techniques, and prevention strategies by referring to studies and International Classification of Diseases, 11th revision

Baseline	**Diagnosis**	**Techniques**	**Prevention strategies**
Age, y	Small papilla	Therapeutic ERCP	Rectal nonsteroidal anti-inflammatory drugs
Sex, female	Fistula of gallbladder or bile duct	Experience of the operator	Pancreatic duct stent placement
Body mass index	Bile duct sludge	Needle-knife sphincterotomy or precut sphincterotomy	Periprocedural fluid solutions
Previous acute pancreatitis	Normal bile duct stone	Transpancreatic sphincterotomy	
Active smoking	Multiple bile duct stones	Difficult cannulation	
Surgically altered anatomy	Huge bile duct stones (>10 mm)	Procedure time >20 min	
Dilated bile duct (6 mm)	Impacted bile duct stones	Endoscopic Papillary Balloon Dilatation (≤10 mm)	
Normal bilirubin level (1.2 mg/dL)	Intrahepatic cholelithiasis	Endoscopic Papillary Large Balloon Dilatation (>12 mm)	
	Calculus of gallbladder or cystic duct	Bile duct stent	
	Mirizzi syndrome	Mechanical lithotripsy	
	Biliary stricture	Endoscopic sphincterotomy	
	Pancreatic duct stricture	Endoscopic retrograde biliary drainage	
	Dilated pancreatic duct	Endoscopic nasobiliary drainage	
	Occupied lesion (cystic diseases or neoplasms of pancreas)	Endoscopic nasopancreatic drainage	
	Intraoperative bleeding	Biopsy sampling	
	Cholangitis	Indwelling T-tube	
	Cholecystitis		
	Diverticulum of duodenum		
	Choledochopancreatic junction anomaly		
	Sphincter of Oddi disorders		
	Duodenal ulcer		
	Specified acquired anatomic alterations of gallbladder or bile ducts		

EHR, Electronic health record.

SUPPLEMENTARY TABLE 3. Comparison of demographic characteristics of patients from hospital 1 (internal validation) with hospitals 2 and 3 (external validation) using analysis of variance and the $\chi 2$ test for continuous and categorical variables, respectively

	Internal validation		Internal total	External validation
	Hospital 1		Hospital 1	Hospitals 2 and 3
Demographics	Training with 10-fold cross-validation dataset (70%)	Testing dataset (30%)	Training and testing	Testing dataset
Mean age, y (standard deviation)	62.16 (15.06)	62.71 (15.19)	62.33 (15.10)	64.25 (14.53)
Sex				
Male	575 (48.16)	247 (48.24)	822 (48.18)	103 (49.05)
Female	619 (51.84)	265 (51.76)	884 (51.82)	107 (50.95)
Post-ERCP pancreatitis	82 (6.87)	36 (7.03)	118 (6.92)	15 (7.14)
No. of major duodenal papilla images	5374	2098	7472	668
No. of cases	1194	512	1706	210
No. of patients	1168	508	1676	210

Values are n (%) unless otherwise defined.

SUPPLEMENTARY TABLE 4. Comparison between previous studies and our study

Study	Aims	Design of study	Subjects	PEP prediction			Our study	
				Validation	Input	AUC	AUC	Justification
Zheng et al[26] (2020)	To develop and validate a predictive model for the occurrence of PEP with factors that are assessed routinely.	Retrospective, multicenter machine learning study	2043 patients with native papilla underwent ERCP for training set, 612 patients for internal test set, 342 patients for external test set. PEP rate: 13.7% (47/342), 7.2% (148/2043)	Internal validation	Patient-related data Procedure-related data	.73	.96	1. Our models predicted PEP in routine ERCP group, including characteristics that were routinely collected, which was suitable for a broader patient population. 2. Our study first proposed the need for evaluating macroscopic shape of the major duodenal papilla in PEP prediction. 3. Our traditional logistic regression model exceeded or was at least on par with the performance of other previous studies (AUC = .86). In contrast to the logistic regression model, the random forest model achieved an AUC of .96 using multimodal data including major duodenal papilla images and electronic health record data.
				External validation		.72	.99	
Fu et al[24] (2023)	To develop and validate a predictive model for PEP, particularly for patients after biliary stent placement because of malignant biliary obstruction.	Retrospective, single-center logistic regression study	1524 patients were enrolled, including 1016 in the development cohort and 508 in the validation cohort, with an overall PEP rate of 7.1%.	Internal Validation	Acute pancreatitis history, the absence of pancreatic duct dilation, nonpancreatic cancer, difficult cannulation, and pancreatic injection.	.78	NA	
Livia et al (2022) Machine learning for the prediction of post-ERCP pancreatitis risk: a proof-of-concept study	To build and evaluate the performance of machine learning models to predict PEP probability and identify relevant features.	Prospective, multicenter machine learning contrast study	1150 patients were included, and 6.1% developed PEP.	Internal validation	Bilirubin, age, body mass index, procedure time, previous sphincterotomy, alcohol units per day, cannulation attempts, sex, gallstones, use of Ringer solution, and periprocedural nonsteroidal anti-inflammatory drug use.	.61	NA	
Chan et al (2021) Pre- and post-procedure risk prediction models for post-ERCP pancreatitis	To develop and validate a risk prediction model for PEP using the latest clinical database.	Retrospective, single-center logistic regression study	1495 patients with an overall PEP rate of 4.6%. 760 and 735 patients in the derivation and validation cohorts, respectively.	Internal validation	Age ≤65 y, female sex, acute pancreatitis history, malignant biliary obstruction, and pancreatic sphincterotomy were independent risk factors.	NA	NA	
Masafumi et al (2020) The milestone for preventing post-ERCP pancreatitis using novel simplified predictive scoring system: a propensity score analysis	To establish a simplified predictive scoring system for PEP.	Retrospective, single-center propensity score analyses	3362 consecutive transpapillary ERCP procedures	Internal validation	Patient-related data and procedure-related data were extracted from the medical charts, based on a review of the risk factors of PEP.	.86	NA	

(*continued on the next page*)

SUPPLEMENTARY TABLE 4. Continued

Study	Aims	Design of study	Subjects	PEP prediction			Our study	
				Validation	Input	AUC	AUC	Justification
Gregory et al (2015) Development and validation of a prediction model for admission following endoscopic retrograde cholangiopancreatography	To develop a model that would allow physicians to predict outcomes of patients during postanesthesia recovery.	Retrospective, single-center logistic regression study	3424 ERCPs during the study period and 3.7% developed PEP.	Internal validation	Procedure indication and complexity, along with female sex, baseline comorbidities, and pain characteristics during the first hour in postanesthesia care unit were the most important factors associated with unplanned admission.	.78	NA	
Matthew (2013) Risk models for post endoscopic retrograde cholangiopancreatography pancreatitis (PEP)	To investigate which variables were independently associated with PEP and derive predictive risk models for PEP.	Retrospective, single-center logistic regression study	6505 patients had 8264 ERCPs, 211 patients had PEP, and 22 patients had severe PEP.	Internal validation	Protective (current smoking, chronic liver disease, chronic liver disease adverse events). Predictive: younger age, suspected sphincter of Oddi disorders, pancreatic sphincterotomy, moderate to difficult cannulation.	NA		

AUC, Area under the receiver-operating characteristic curve; *PEP*, post-ERCP pancreatitis; *NA*, not available.